

Digital Agro-Meteorology and Index-Based Micro-Insurance for Smallholder Climate Resilience in Nsukka, Nigeria: A Transdisciplinary Pilot for the Guinea-Savanna Transition Zone

Mabrig Korie

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Author Affiliation

Mabrig Korie

Mabrig Research Institute, Lagos, Nigeria

Email: contact@mabrigkorie.org

Website: <https://www.mabrigresearch.online/>

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Abstract

Smallholder farmers in the Nsukka agricultural zone of Enugu State, Southeast Nigeria, stand at the sharp edge of climate disruption. They grow the crops that feed families and sustain local markets—yams, cassava, and the distinctive Nsukka yellow pepper—but the rains no longer behave as they once did. False starts to the planting season, punishing dry spells in the middle of what should be steady growth, and late-season downpours that cut short maturation have become the new normal. National meteorological forecasts, designed for broad regional scales, offer little guidance for the steep micro-climates created by the Nsukka-Udi hills. When crops fail, there is no reliable safety net. Traditional agricultural insurance remains out of reach for most rural households because of high costs, slow claims processes, and the simple absence of verifiable field-level data.

This eighteen-month transdisciplinary pilot project sets out to change that equation. We will deploy a network of ten low-cost, solar-powered Internet of Things weather stations across the University of Nigeria Nsukka research fields and three neighbouring communities—Obukpa, Alor-Uno, and Eha-Alumona. These stations will record rainfall, air and soil temperature, and soil moisture at fifteen-minute intervals, feeding real-time

data into an open cloud platform. Historical satellite records from CHIRPS and MODIS will be fused with this ground truth to produce fine-grained micro-climate maps that actually reflect what farmers experience on different slopes and aspects.

That environmental intelligence will then drive two practical tools. First, an automated advisory system will translate sensor readings into plain-language planting, fertiliser, and harvesting guidance delivered straight to farmers' basic feature phones through USSD and SMS, using Nigeria's existing mobile infrastructure and local languages. Second, the same sensor network will underpin a parametric micro-insurance product. Instead of sending assessors to inspect damaged fields after every shock, the system will trigger automatic payouts when consecutive dry days exceed a pre-agreed threshold during critical growth windows. Payouts will arrive via mobile money within days, not months, giving households cash precisely when they need it most to replant or buy food.

The project marries engineering, agronomy, development economics, and community knowledge. Two hundred smallholder households will participate in baseline surveys and focus-group discussions that ground the technical design in lived reality and reveal willingness-to-pay thresholds. All data will be governed by Nigeria's Data Protection Regulation, with clear benefit-sharing agreements so that the communities themselves retain ownership and long-term access. By the end of the pilot we expect three concrete deliverables: a public Nsukka micro-climate dashboard hosted at the university, a validated digital crop-weather calendar tailored to current variability, and an investment-ready parametric insurance model that can be scaled through microfinance institutions, agricultural development banks, and emerging climate-finance facilities.

What we are proposing is not another top-down technology transfer. It is a locally anchored system that turns granular environmental data into both better day-to-day decisions and a genuine financial cushion. If successful, the model can be adapted across the Guinea-Savanna belt and other rain-fed smallholder landscapes where conventional insurance and coarse forecasts have consistently failed the people who need them most.

Keywords: smallholder resilience, index-based insurance, agro-meteorology, IoT weather stations, climate adaptation, Nsukka yellow pepper, Southeast Nigeria, parametric triggers, USSD advisory systems, micro-climate mapping

1. Introduction

Walk through the red-earth paths of Obukpa or Eha-Alumona any April and you will still see farmers watching the sky with the same intensity their grandparents did. The first heavy showers used to mean one thing: it was time to plant. Today those same showers often arrive, tempt everyone into the fields, and then disappear for three or four weeks, leaving young yam setts and pepper seedlings to wither under a suddenly fierce sun. This is not anecdotal grumbling. It is the lived experience of thousands of smallholder households whose entire annual income and food security hinge on rainfall patterns that have become markedly less predictable over the past two decades.

Nsukka sits in the Guinea-Savanna transition zone of Southeast Nigeria, a landscape of rolling hills and shallow valleys where elevation changes of only a few hundred metres create surprisingly sharp differences in temperature, humidity, and the timing of rain. The University of Nigeria Nsukka has long been a centre for agricultural research precisely because of this variability. Yet the very features that make the area scientifically interesting also make conventional weather information almost useless for operational farming decisions. The Nigerian Meteorological Agency produces excellent synoptic forecasts, but these are calibrated for airports and large administrative regions. A forecast that says “Enugu State will receive normal rainfall” tells a farmer on the windward slope of a Nsukka hill almost nothing about whether her particular plot will stay moist enough for the next ten days.

The crops at stake are not minor. Yam remains the cultural and calorific anchor of Igbo farming systems. Cassava provides the reliable fallback that smooths over bad yam years. And Nsukka yellow pepper—*Capsicum chinense*—is a high-value niche crop whose distinctive aroma and colour command premium prices in urban markets across Southern Nigeria and beyond. When any one of these fails, the ripple effects move quickly from household food stocks to school fees, medical bills, and the ability to buy seed for the next season.

Climate change is not a distant threat here; it is already rewriting the agricultural calendar. Studies across Southeast Nigeria document delayed onset of rains, more frequent mid-season dry spells, increased intensity of individual storms, and occasional acid-rain events linked to harmattan dust and industrial emissions. The statistical relationships are clear: each additional week of delayed planting correlates with measurable yield declines in yam and pepper. What the numbers cannot capture is the quiet desperation that sets in when a family has already invested labour and borrowed money for seed and fertiliser, only to watch the seedlings die.

Existing institutional responses have not kept pace. The Nigerian Agricultural Insurance Corporation offers

indemnity-based products, but the paperwork, field verification visits, and delays in payout make them effectively inaccessible to anyone farming less than five hectares—and most Nsukka smallholders operate on one or two. Index-based insurance, which pays out on the basis of an objective weather index rather than individual loss assessment, has shown promise elsewhere in Africa. Kenya's Kilimo Salama programme demonstrated that weather-station or satellite triggers, combined with mobile-money delivery, can reach tens of thousands of smallholders at premiums they can actually afford. Yet those successes have not travelled to Southeast Nigeria, partly because the micro-climate granularity required simply does not exist in the available data layers.

This project begins from a simple conviction: if smallholders are to adapt rather than merely survive, they need two things at once. They need information that is accurate at the scale of their fields and delivered in time for real decisions. And they need a financial instrument that converts that same information into rapid, automatic liquidity when the worst happens. Digital agro-meteorology and parametric micro-insurance are not separate solutions; they are two sides of the same data infrastructure.

The pages that follow lay out a practical, eighteen-month pilot designed to build exactly that infrastructure in Nsukka. We describe the sensor network, the community engagement process, the advisory and insurance architectures, the expected deliverables, and the pathway to scale. Throughout we have tried to keep one test in mind: would a farmer in Alor-Uno, looking at her phone in the middle of a dry spell, actually change what she does because of this system? If the answer is yes, we will have succeeded. If not, we will have built another elegant pilot that changes nothing on the ground.

2. Statement of the Problem

The fundamental problem is not that it rains too much or too little in Nsukka. The problem is that no one can say with any confidence, at the beginning of a given week in April or May, whether the next ten days will bring the steady moisture a yam crop needs or a desiccating dry spell. That uncertainty is now structural.

Historical rainfall records for the zone show a clear shift. The long-term mean onset of the rainy season has moved later by roughly ten to fourteen days compared with the 1980–2000 baseline. More critically, the variance around that mean has increased. A farmer who plants on the basis of the old “first heavy rain after the twentieth of March” rule now faces a coin-flip probability that the follow-up rains will arrive within the critical germination window. When they do not, the cost is not merely lower yield; it is the loss of an entire season's investment and the debt that often accompanies it.

The Nigerian Meteorological Agency's public forecasts remain valuable for aviation, flood warning, and broad agricultural planning at state level. They are simply not built for the micro-climate reality of Nsukka's dissected topography. A single synoptic station cannot capture the rain-shadow effects that can leave one village with adequate moisture while another three kilometres away, on the opposite slope, is already cracking dry. Satellite products such as CHIRPS improve spatial coverage but still smooth over the very local gradients that determine whether a pepper seedling survives its first fortnight.

Compounding the information gap is the complete absence of any functional risk-transfer mechanism. When a dry spell or an early cessation of rains destroys a crop, the household absorbs the entire loss. There is no insurance payout, no government relief that arrives in time to matter, and only limited informal borrowing from neighbours who are often in the same situation. The result is a slow but steady erosion of household assets—sale of livestock, withdrawal of children from school, migration of young adults—and a deepening reluctance to invest in yield-enhancing inputs because the downside risk feels unmanageable.

Traditional indemnity insurance cannot solve this at smallholder scale. The transaction costs of sending an assessor to every claim, the moral-hazard problems, and the simple lack of historical yield data at plot level make premiums either unaffordable or the products unprofitable for insurers. Parametric or index-based products remove the need for individual loss adjustment by paying out whenever an objective, verifiable index crosses a threshold. The challenge in Nsukka has been the lack of an index that is both accurate enough to minimise basis risk (the mismatch between index trigger and actual loss) and cheap enough to maintain at the density required by rolling terrain.

Without a dense, real-time agro-meteorological observation network feeding both advisory services and parametric triggers, smallholders in Nsukka will remain trapped in a cycle of autonomous, low-investment, high-vulnerability farming. Each successive climate shock will push more households below the threshold of viable reproduction, accelerating land degradation as desperate families mine their soils for whatever short-term yield remains. The window for building the necessary data infrastructure before these dynamics become irreversible is narrowing.

3. Literature Review

The scholarly and practical literature on climate adaptation in African smallholder systems has grown rapidly, yet it still contains a stubborn gap between macro-level analysis and the micro-level tools farmers actually need.

A substantial body of work documents the biophysical impacts of climate variability on staple crops across West and Central Africa. Ezeh and colleagues, working in Southeast Nigeria, found statistically significant negative relationships between delayed rainfall onset, mid-season dry spells, and yields of yam, cassava, and pepper. Their models suggest that each additional week of delay in planting can reduce yam output by measurable percentages, with effects compounded by acid-rain events that are increasingly reported in the region. Similar patterns appear in studies from the savanna zones of Ghana, Burkina Faso, and northern Nigeria: the timing of rainfall matters more than total seasonal volume for many rain-fed systems.

On the institutional side, index-based weather insurance has moved from pilot curiosity to established instrument in several African countries. Kenya's Kilimo Salama programme, launched in 2009 by Syngenta Foundation, UAP Insurance, and Safaricom, used a network of automated weather stations to trigger payouts for drought or excess rain on maize and other crops. By 2015 the product had reached more than 200,000 farmers, with payouts delivered via M-Pesa within forty-eight hours of a trigger. Independent evaluations showed reduced distress sales of assets and modest increases in input use among insured households, although uptake remained sensitive to premium levels and trust in the index. Ghana's Agricultural Insurance Programme and Ethiopia's R4 Rural Resilience Initiative have demonstrated comparable logic using satellite rainfall estimates, achieving scale through integration with existing social-protection or credit systems.

Yet these successes rest on two conditions that are not automatically present everywhere. First, the index must be dense enough or well enough calibrated that basis risk—the gap between what the index says and what the farmer actually experiences—stays within tolerable bounds. In landscapes with steep micro-climate gradients, a single weather station or coarse satellite pixel can easily misclassify a farm's experience. Second, the delivery infrastructure (mobile money, agent networks, farmer organisations) must already be strong enough to handle rapid, low-friction payouts and premium collection. Where these conditions are only partially met, pilots have struggled with low renewal rates and persistent distrust.

Nigeria's own experience with agricultural insurance has been dominated by the Nigerian Agricultural Insurance Corporation's indemnity products. While NAIC has expanded its portfolio, penetration among smallholders remains negligible. Administrative costs, the requirement for field verification, and the absence of granular weather or yield data have kept products either subsidised at unsustainable levels or priced beyond the reach of the very farmers they are meant to serve. Recent policy interest in climate-smart agriculture and the National Climate Change Action Plan has created rhetorical space for innovative instruments, but concrete implementation at scale has lagged.

Parallel to the insurance literature, a growing body of work explores the potential of low-cost Internet of

Things technologies for agricultural monitoring in low-resource settings. Projects such as the Trans-African Hydro-Meteorological Observatory (TAHMO) have deployed hundreds of robust, low-maintenance stations across the continent, proving that reliable data collection need not cost tens of thousands of dollars per site. Open-source platforms and GSM telemetry modules now allow real-time transmission even in areas with patchy power and network coverage. The missing link has been the integration of such sensor networks with both advisory systems that speak the language of farmers and financial instruments that convert the same data into automatic liquidity.

Community-based and participatory approaches to climate information services have also gained traction. When farmers are involved in defining what counts as a “bad” season, in validating sensor readings against their own observations, and in co-designing the format and timing of advisories, both trust and utilisation rise. Conversely, purely top-down SMS blast systems have often seen low open rates and even lower behavioural change. The literature therefore points toward hybrid models that combine objective sensor data with local knowledge validation and two-way feedback loops.

What remains strikingly absent is a documented case in Southeast Nigeria—or indeed in much of the humid Guinea-Savanna zone—where a dense agro-meteorological network has been deliberately designed to feed both operational advisories and a parametric insurance trigger. Most existing index products in West Africa still rely on satellite rainfall estimates at 5–10 km resolution or on widely spaced synoptic stations. For Nsukka’s topography, that resolution is simply too coarse. The present project is conceived as a deliberate attempt to close that specific gap: to build, at pilot scale, the missing micro-climate data layer and to hard-wire it into both decision-support and risk-transfer tools that smallholders can actually use.

4. Research Objectives

The overarching aim is to co-design and pilot a micro-climate-informed advisory and parametric insurance system that measurably strengthens the climate resilience of smallholder households in the Nsukka agricultural zone.

Specific objectives are:

1. To design, deploy, and validate a network of ten low-cost IoT weather stations that provide continuous, fifteen-minute resolution data on rainfall, temperature, and soil moisture across representative slope positions and land uses in Obukpa, Alor-Uno, Eha-Alumona, and the University of Nigeria Nsukka research fields.

2. To fuse real-time sensor observations with historical CHIRPS and MODIS archives to produce validated micro-climate maps and a dynamic crop-weather calendar that reflects current rainfall variability rather than outdated normals.
3. To develop and test an automated USSD/SMS advisory engine that translates sensor-derived soil-moisture and rainfall metrics into timely, plain-language guidance on planting windows, fertiliser timing, and harvest decisions, delivered in Igbo and Nigerian Pidgin to feature-phone users.
4. To construct and calibrate a parametric insurance trigger based on consecutive dry days during critical vegetative growth phases, and to model premium and payout structures that are both actuarially sound and affordable to smallholder households.
5. To document smallholder willingness-to-pay, perceived basis risk, and behavioural responses through a mixed-methods evaluation involving 200 surveyed households and repeated focus-group validation, ensuring that the final system design is grounded in local priorities and constraints.
6. To produce three public goods—an open Nsukka micro-climate data dashboard, a replicable digital crop-weather calendar, and an investment-ready parametric insurance white paper—that can be taken up by the University of Nigeria, state agricultural development programmes, microfinance institutions, and climate-finance facilities for scaling beyond the pilot phase.

5. Methodology

The project adopts a transdisciplinary, mixed-methods design that deliberately weaves together sensor engineering, agronomic modelling, development economics, and participatory community research. Implementation is organised into four sequential but overlapping phases over eighteen months.

Phase 1: Micro-Climate Sensor Deployment and Grid Design

The study area encompasses the University of Nigeria Nsukka agricultural research station and three contiguous agrarian communities—Obukpa, Alor-Uno, and Eha-Alumona—chosen to capture both valley-bottom and upper-slope positions as well as a gradient of market access and previous exposure to extension services.

Each of the ten weather stations will be built around an ESP32 microcontroller with solar panel, 10,000 mAh lithium battery, and GSM telemetry module. Sensors include a tipping-bucket rain gauge (0.2 mm resolution), digital air temperature and humidity probe, and capacitive soil-moisture sensors installed at 10 cm and 30

cm depths. All components are selected for robustness under tropical conditions and sourced from suppliers with proven field performance in sub-Saharan deployments. Total hardware cost per station is budgeted at approximately \$180, including assembly and initial calibration.

Stations will be sited in consultation with community leaders and university agronomists to represent the dominant land-use types (yam/cassava intercropped with pepper, sole pepper plots, fallow) and topographic positions. Data are sampled every fifteen minutes and transmitted via GSM to a MongoDB database hosted on a Render cloud instance. A lightweight Python/FastAPI backend performs quality control, gap-filling using nearby stations and satellite estimates, and exposes both raw and processed data through a public API.

Historical context will be added by ingesting CHIRPS daily rainfall (1981–present) and MODIS land-surface temperature and vegetation indices at 1 km and 250 m resolution. These layers will be downscaled and bias-corrected against the new ground observations to produce a seamless twenty-year micro-climate time series for each station location. The resulting maps will reveal, for the first time, the actual spatial structure of rainfall and temperature anomalies across Nsukka’s hills—information that has never been available at this resolution for operational use.

Phase 2: Agrarian Baseline and Socio-Economic Field Surveys

Technical data alone cannot design a useful service. We therefore embed a substantial social-science component from the outset.

A multi-stage stratified random sample of 200 smallholder households will be drawn from the three communities. Stratification variables include primary crop emphasis (yam-dominant, cassava-dominant, Nsukka yellow pepper specialists) and gender of the principal decision-maker, ensuring that women’s knowledge and constraints are not marginalised. Sample size is powered to detect moderate differences in willingness-to-pay and perceived climate risk across strata.

Enumerators recruited from the local area and trained by the research team will administer a structured digital questionnaire using ODK Collect on Android tablets. The instrument captures: household demographics and asset portfolios; historical planting and harvesting calendars; self-reported yield losses attributed to specific weather events over the past five seasons; current coping strategies (informal borrowing, asset sales, off-farm labour, reduced consumption); and stated willingness-to-pay for an index-based insurance product under different premium and payout scenarios. A separate module elicits farmers’ own definitions of a “bad” season and their preferred format and timing for weather advisories.

Complementing the survey, eight focus-group discussions (two per community, segregated by gender and age where culturally appropriate) will explore autonomous adaptation practices in depth, surface local indicators of soil moisture and plant stress, and validate or challenge the sensor-derived thresholds we propose to use. These discussions also serve as the first step in building community ownership and feedback channels that will continue throughout the pilot.

All research protocols will be reviewed by the University of Nigeria Nsukka Institutional Review Board. Informed consent will be obtained in Igbo or English according to participant preference, and data handling will comply fully with the Nigeria Data Protection Regulation 2019. Raw household data will be anonymised before any public release; community-level aggregates will be returned to participating villages in accessible formats.

Phase 3: Predictive Matrix Development and USSD Advisory Architecture

Raw sensor numbers become useful only when they are translated into decisions farmers can act on. This phase builds that translation layer.

Using the combined sensor-plus-satellite dataset, we will calculate running indices of soil-moisture depletion and cumulative rainfall over biologically meaningful windows (for example, the first thirty days after planting for yam and pepper). Thresholds will be derived statistically from historical yield-weather relationships reported in the literature and refined through the focus-group validation process. For instance, a cumulative rainfall total below 60 mm in the thirty days after yam planting, or more than eight consecutive days with soil moisture below 15 % volumetric water content during vegetative growth, may be flagged as high-risk.

These thresholds populate a simple decision matrix implemented in JavaScript (Math.js library) on the backend. When incoming data cross a threshold, an automated script generates a short, actionable advisory in plain language. Examples include: “Heavy rain expected in next 3 days. Delay fertiliser application on pepper until soil drains.” or “Soil moisture dropping fast on upper slopes. Consider supplementary watering for young yam if you have access to stream.” Messages are translated into standard Igbo and Nigerian Pidgin by a local language specialist and reviewed by community members for clarity and cultural fit.

Delivery occurs through Africa’s Talking USSD and SMS gateway. USSD is chosen because it works on any GSM phone without internet or smartphone ownership—still the reality for a significant minority of older farmers in the area. The system supports two-way interaction: a farmer can query current conditions for her village or report observed crop stress, feeding additional ground-truth back into the model. All messages are

logged and linked to anonymised household identifiers so that we can later analyse which advisories were received, opened, and acted upon.

Phase 4: Index-Based Micro-Insurance Parametric Modelling

The same sensor network that feeds advisories will underpin the insurance trigger, ensuring perfect alignment between the information farmers receive and the conditions that determine payouts.

After extensive consultation with actuaries, agricultural economists, and the focus groups, we have selected **Consecutive Dry Days (CDD)** during the critical April–June vegetative window as the primary index. A dry day is defined as any day with less than 1 mm of rainfall recorded by the nearest station (or interpolated value). If the network records ten or more consecutive dry days between 1 April and 30 June, the parametric trigger fires for all enrolled farmers whose plots fall within the affected micro-climate zone.

This threshold was chosen because it corresponds to documented stress points for yam and pepper establishment while remaining rare enough under current climatology to keep premiums manageable. Payout amounts will be modelled at two levels: a base payment of ₦15,000 per enrolled hectare (roughly the cost of replanting and basic inputs) and a top-up tier for more severe events. Premiums will be calculated using burn analysis on the twenty-year reconstructed dataset, with conservative loading for uncertainty and operational costs. We anticipate a commercial premium in the range of ₦2,500–4,000 per hectare per season before any subsidy; the pilot will test both fully commercial and partially subsidised scenarios.

Payouts will be executed automatically via mobile money (bank transfer or USSD-enabled wallets) within seventy-two hours of trigger confirmation, with no claims form required. This speed is central to the value proposition: cash arrives while the shock is still unfolding, allowing households to protect consumption, retain productive assets, and in some cases replant a short-cycle crop.

To address basis risk—the inevitable mismatch between any index and individual plot experience—we will incorporate a small “ground-truth top-up” mechanism. A random 10 % sample of triggered claims will receive a rapid field visit; if actual losses demonstrably exceed the index payout, an additional discretionary payment can be authorised from a contingency reserve. While this reintroduces some administrative cost, it provides a visible fairness valve that builds trust in the early years of the product.

The insurance component will be developed in partnership with a Nigerian microfinance institution or insurance company willing to underwrite on a pilot basis, with technical support from the research team on index construction and pricing. All modelling code and assumptions will be fully documented in an open

white paper so that other actors can audit, replicate, or adapt the approach.

Data Management, Quality Assurance, and Ethics

All environmental and household data will be stored in encrypted form on the Render/MongoDB instance with daily automated backups. Access logs will be maintained; only authorised project personnel and designated community data stewards will have download privileges. A public-facing dashboard will display only aggregated, anonymised visualisations and station-level time series, protecting individual farm locations.

Quality assurance includes automated flagging of sensor drift or transmission gaps, cross-validation against neighbouring stations and satellite estimates, and periodic manual calibration checks during field visits. The project will produce a concise Standard Operating Procedure manual so that the system can be maintained or expanded by University of Nigeria technical staff after external funding ends.

Ethical practice is non-negotiable. Beyond formal IRB approval, we will establish a small community advisory committee in each village that meets quarterly to review emerging findings, raise concerns, and co-interpret results before any public dissemination. Data-use agreements will specify that communities retain rights to their own aggregated information and that any commercial use of the insurance model beyond the pilot requires renegotiation of benefit-sharing terms.

6. Implementation Timeline and Indicative Budget

The project runs for eighteen months, divided into an intensive six-month mobilisation and deployment phase, a full twelve-month operational and evaluation phase, and a final two-month synthesis and handover window. A simplified Gantt overview appears below; detailed work plans with milestones are available in supplementary materials.

Months 1–3: Stakeholder mapping, IRB clearance, community entry meetings, recruitment and training of enumerators and local technicians, final sensor hardware procurement and bench testing.

Months 4–6: Station installation and commissioning, baseline survey rollout, focus-group series, initial historical data fusion and threshold calibration, backend and USSD prototype development.

Months 7–15: Full data collection and real-time advisory operation, insurance product finalisation and regulatory engagement, mid-term community feedback workshops, continuous model refinement based on farmer input and sensor performance.

Months 16–18: Endline survey and impact assessment, documentation of lessons, production of dashboard, crop-weather calendar, and insurance white paper, formal handover to University of Nigeria Nsukka and participating communities, policy engagement events with state and federal agricultural agencies.

Indicative Budget (USD, approximate June 2026 prices)

Category	Amount (USD)	Notes
Sensor network (10 stations + spares)	2,800	Hardware, assembly, initial calibration
Cloud infrastructure & SMS gateway	1,200	18 months Render + Africa's Talking usage
Fieldwork & surveys (200 HH)	6,500	Enumerator training, transport, incentives, tablets
Community engagement & FGDs	2,000	Meetings, refreshments, local facilitators
Personnel (lead researcher, 2 assistants, data analyst)	18,000	Part-time over 18 months, local rates
Travel, workshops, dissemination	3,500	Stakeholder events, policy brief printing

Category	Amount (USD)	Notes
Contingency (10 %)	3,400	Unforeseen hardware, currency fluctuation
Total	37,400	

We are actively exploring co-funding from the University of Nigeria research board, Enugu State Ministry of Agriculture, and international climate-adaptation facilities. In-kind contributions from UNN (lab space, student interns, existing meteorological mast access) will further reduce cash requirements.

7. Expected Grant Deliverables and Policy Value

By the close of the eighteen-month pilot, three tangible public goods will be in community and institutional hands.

The Nsukka Micro-Climate Open Data Dashboard. A live, publicly accessible web interface displaying real-time readings from all ten stations, historical time-series graphs, and downloadable micro-climate maps. Hosted initially on the project Render instance and mirrored at the University of Nigeria Nsukka Department of Soil Science or Agricultural Engineering, the dashboard will serve researchers, extension officers, and—crucially—any farmer or cooperative with smartphone access who wants to see conditions beyond their own farm gate. Long-term maintenance will be absorbed into university teaching and research budgets.

The Localised Digital Crop-Weather Calendar. A concise, farmer-friendly booklet and companion digital file that translates the new micro-climate understanding into practical guidance: recommended planting windows under current rainfall variability, soil-moisture thresholds for top-dressing decisions, and early-warning indicators for dry-spell risk. The calendar will be produced in Igbo, English, and Pidgin, tested with focus groups, and distributed through participating cooperatives, churches, and extension agents. Because it is built on continuously updated sensor data rather than static historical normals, it can be revised annually at minimal cost.

The Parametric Micro-Insurance White Paper and Model. A fully documented, investment-ready package

containing: the calibrated CDD trigger algorithm and its statistical performance against historical yield proxies; premium and payout tables under alternative subsidy scenarios; legal and regulatory considerations for Nigerian insurance licensing; sample policy wording; and an actuarial memorandum suitable for presentation to microfinance banks, NAIC, and international climate-finance vehicles such as the Green Climate Fund or African Risk Capacity. The white paper will be open-access and accompanied by anonymised code repositories so that other Nigerian or West African teams can adapt the approach without starting from zero.

Beyond these concrete outputs, the project will generate policy-relevant evidence on three broader questions. First, how much basis risk remains when a dense ground-sensor network is used instead of satellite-only indices, and what residual risk-mitigation mechanisms (top-up payments, area-yield hybrids) are most cost-effective. Second, whether the combination of timely advisories and rapid parametric payouts produces measurable behavioural change—higher input use, greater willingness to plant higher-value crops, reduced distress asset sales—compared with information-only or insurance-only interventions. Third, what institutional arrangements (university-hosted data trust, community data cooperative, private insurer partnership) best ensure long-term sustainability and equitable benefit distribution after donor funding ends.

These lessons will be packaged into policy briefs targeted at the Enugu State Ministry of Agriculture and Rural Development, the Federal Ministry of Agriculture and Food Security's Climate-Smart Agriculture unit, and the growing community of Nigerian and African climate-finance practitioners. We will also seek opportunities to present findings at regional conferences (e.g., African Climate-Smart Agriculture Alliance) and to contribute to national dialogues on agricultural insurance reform.

8. Risks, Limitations, and Mitigation Strategies

No pilot of this complexity is risk-free. We have identified the principal risks and built explicit mitigation measures into the design.

Technical risk: Sensor failure or GSM network outages could create data gaps. Mitigation: redundant power design, local spare-parts stock, automated quality-control scripts that interpolate from neighbours and satellite data, and a maintenance protocol that trains university technicians and one community member per site in basic troubleshooting.

Basis-risk and trust risk: If the index triggers when farmers have not suffered major loss (or fails to trigger when they have), confidence will erode quickly. Mitigation: conservative threshold setting informed by both statistics and farmer testimony; transparent communication of how the index works; the 10 % ground-truth

top-up mechanism; and continuous feedback loops through USSD and quarterly community meetings.

Low uptake or willingness-to-pay: Even a well-designed product may see limited voluntary enrolment if premiums feel high or the value proposition is not yet trusted. Mitigation: the pilot will test both fully paid and heavily subsidised premium scenarios; integration with existing cooperative or credit-group structures can bundle insurance with input loans; and the advisory service provides immediate, tangible value even before any payout occurs, building habit and trust.

Institutional and regulatory risk: Securing a willing insurance underwriter and navigating Central Bank of Nigeria or NAICOM rules for parametric products may take longer than planned. Mitigation: early engagement with NAIC and at least two microfinance institutions; production of a regulatory-options memo in Month 4; willingness to begin with a mutual or community-based risk pool if formal underwriting is delayed.

Sustainability risk: After eighteen months the system must not collapse when external funding ends. Mitigation: deliberate capacity transfer to UNN (student involvement, thesis linkages, technical documentation); community data-steward training; design of a low-cost maintenance budget that can be absorbed by university research overhead or small user fees for premium dashboard features; and explicit exit strategy agreed with community advisory committees from Month 3 onward.

9. Conclusion and Recommendations

The smallholder farmers of Nsukka do not need another report telling them that the climate is changing. They need information they can act on before the next planting window closes, and they need a financial backstop that does not require them to prove their poverty to a distant claims officer after the damage is done. This project proposes to deliver both, using the same underlying data infrastructure: a dense, low-cost sensor network whose readings feed both real-time agronomic advisories and an objective parametric insurance trigger.

By grounding every design choice in the lived experience of two hundred households and by embedding the entire system within existing university and community institutions, we aim to produce not a showcase pilot that ends when the funding does, but a living platform that continues to generate value and can be extended—first across Enugu State and then to comparable agro-ecologies in the broader Guinea-Savanna transition.

We therefore recommend that climate-adaptation and agricultural-development funders support this pilot at

the scale outlined. The investment is modest relative to the potential returns in protected household assets, stabilised food production, and a replicable model for other Nigerian and West African smallholder landscapes where conventional approaches have reached their limits. More importantly, it treats smallholders not as passive recipients of technology or aid, but as knowledgeable partners whose daily decisions, aggregated across thousands of farms, will ultimately determine whether Nigeria's agrarian heartlands thrive or merely endure under twenty-first-century climate conditions.

The rains will continue to arrive and depart according to their own logic. What we can change is whether those rhythms still catch farmers unprepared, or whether they are met with eyes open, phones in hand, and a modest but reliable financial cushion ready to deploy when the dry spell stretches one week too long.

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Conflict of Interest

The author declares no conflict of interest.

Data Availability

Anonymised micro-climate sensor data, survey instruments, and modelling code will be made openly available via the Mabrig Research Institute repository (<https://www.mabrigresearch.online/>) upon pilot completion or upon reasonable request.

Supplementary Materials

Detailed budget spreadsheet, sensor deployment and calibration protocols (including soil-specific thermogravimetric methods), sample USSD advisory templates, and statistical annexes are available as online supplementary files linked to this article.

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